# GeoFusionNet / GeoCorr: Status Report

Status as of 2026-10-11, 20:40 UTC (02:10 IST 10-12) · main @ edf0765 · Prepared for Gobika S

> **Conventions.** DTU test, 22 scans, 1600×1152, official protocol. Values are in mm; lower is better. Every "±" in this report is the population standard deviation over seeds (**ddof = 0**), for A0f and C1 alike. **PENDING** marks a number that has not been measured yet.

## 1. Status

| Item | State | Expected |
|---|---|---|
| **C1 3-seed eval** (B2_capacity s0, s1, s2) | **Done** 2026-10-11 07:15 UTC | n/a |
| **Track 1 screen** (B2_mono_feat s2) | **Done** 2026-10-11 11:07 UTC | n/a |
| **Fusion sweep, full grid** (A0f s2 vs C1 s2)<br>scripts/queue_fusion_sweep.sh · runs/fusion_sweep/ | **Stopped** 2026-10-11 18:17 UTC after 6 of 44 model-scans (scan1, scan4, scan9 × 2 models); partial outputs kept. It ran at ~3.3–4 h per model-scan (projected end ~10-12 23:30 IST) and slowed training by ~2.4%. | n/a |
| **Fusion sweep, reduced** (distance only, n = 4)<br>scripts/queue_fusion_sweep_dist_n4.sh · runs/fusion_sweep_dist_n4/ | **Done** 2026-10-11 20:35 UTC; see section 3 | n/a |
| **B2_nogeo seed 0** (L_geo off, full chain)<br>scripts/queue_b2_nogeo_seed0.sh · log runs/queue_b2_nogeo_seed0.log | Base training, step 10000/20000 at 18:17 UTC (2.10 s/step vs 2.05 on seed 2); then finetune 6k → final 1.5k → diagnostics → 1600×1152 eval (23 h 27 m on seed 2) | **~2026-10-12 17:15–18:30 IST (PENDING)** |
| C1 placebo prior (E) | Prepared, **not launched** (by decision) | n/a |

If B2_nogeo seed 0 has not been evaluated by 20:00 IST on 2026-10-12, this report is finalised with L_geo at n = 1.

## 2. Headline: C1 vs Stage 0 (3 seeds each)

| Model | Acc | Comp | Overall |
|---|---|---|---|
| A0f seed 0 | 0.436 | 0.558 | 0.497 |
| A0f seed 1 | 0.441 | 0.555 | 0.498 |
| A0f seed 2 | 0.435 | 0.524 | 0.479 |
| **A0f (Stage 0), mean ± std (ddof=0)** | 0.437 ± 0.003 | 0.545 ± 0.016 | **0.491 ± 0.009** |
| C1 seed 0 | 0.470 | 0.358 | 0.414 |
| C1 seed 1 | 0.487 | 0.373 | 0.430 |
| C1 seed 2 | 0.474 | 0.362 | 0.418 |
| **C1 (B2_capacity), mean ± std (ddof=0)** | 0.477 ± 0.008 | 0.364 ± 0.006 | **0.421 ± 0.007** |
| C1 − A0f | +0.040 | −0.181 | −0.071 |

C1 is better overall by 0.071 mm, about 14%, and the seed ranges do not overlap (C1 0.414–0.430, A0f 0.479–0.498). This is a trade, not a uniform gain. Completeness improves by 0.18 mm and accuracy gets 0.04 mm worse. The worst scans are the same as A0f's: scan48 (0.71–0.77), scan77, scan11, scan13 and scan49. On seed 2, C1 beats A0f on 17 of 22 scans. CL-MVSNet is not compared here. Our 0.438 number for CL-MVSNet comes from point clouds generated locally at 640×480, and the cause of its gap to the published 0.329 is unconfirmed.

## 3. Does C1's completeness gain survive a looser or stricter fusion? (reduced fusion sweep)

Both models' depth maps (seed 2 each, already dumped at 1600×1152) are re-fused and scored with the official metric. The full 32-setting grid was too slow (~3.3–4 h per model-scan), so it was stopped and replaced by a reduced sweep.

**Rule, fixed before any result was looked at:** all 22 scans; family *geo* (world-distance check) at distance 0.1 / **0.2** / 0.4 / 0.8 mm; consistent-view count fixed at the official 4; prob 0.3. That is the official setting (0.2 mm), one stricter and two looser. Estimated ~30–35 min per model-scan, 38 to run.

> **What this sweep can support:** whether C1's completeness gain and accuracy loss over A0f depend on the fusion *distance* threshold, and a matched-point-count comparison along that one axis, for scans whose point counts overlap.
> **What it cannot support:** any conclusion about the view-count threshold (2/3/6 views not run); any conclusion about MVSNet-style reprojection/relative-depth filters (not run); anything about the gap to CL-MVSNet (different checkpoint and resolution, not part of this sweep). It is single-seed (seed 2 of each model).

| Check | A0f s2 | C1 s2 | Note |
|---|---|---|---|
| Reproduction of the official setting (geo 0.2 mm, 4 views) against eval_1600x1152.json, max \|ΔOverall\| over all 22 scans | 0.0047 | 0.0009 | Small but not exactly 0; well below the differences discussed here |
| Fused points, official setting (mean, M) | 22.05 | 22.34 | C1 fuses ~1% more points |
| Comp at A0f's point count | 0.524 | 0.370 | C1 better on 22/22 scans (22/22 in range) |
| Comp at C1's point count | 0.511 | 0.362 | C1 better on 22/22 scans (22/22 in range) |
| Acc at A0f's / C1's point count | 0.435 / 0.449 | 0.471 / 0.473 | C1 better on 3/22 and 5/22 scans |
| Overall at A0f's / C1's point count | 0.480 / 0.480 | 0.420 / 0.418 | C1 better on 16/22 and 17/22 scans |

### Per-setting means over 22 scans (geo, 4 views)

| Distance | A0f Mpts | A0f Acc | A0f Comp | A0f Overall | C1 Mpts | C1 Acc | C1 Comp | C1 Overall |
|---|---|---|---|---|---|---|---|---|
| 0.1 mm | 13.33 | 0.364 | 0.703 | 0.534 | 13.76 | 0.376 | 0.483 | 0.429 |
| **0.2 mm (official)** | 22.05 | 0.435 | 0.524 | 0.480 | 22.34 | 0.473 | 0.362 | 0.418 |
| 0.4 mm | 29.70 | 0.544 | 0.410 | 0.477 | 30.32 | 0.620 | 0.293 | 0.457 |
| 0.8 mm | 35.82 | 0.714 | 0.338 | 0.526 | 37.12 | 0.838 | 0.257 | 0.547 |

**Verdict (distance threshold only, seed 2 only):** C1's completeness gain is not an artefact of the distance threshold. At matched point counts C1 has better completeness on every one of the 22 scans, and its accuracy is worse on most scans (C1 better on only 3–5 of 22). So the trade-off is a property of the depth maps, not of how many points the fusion keeps. C1 is better overall at 0.1, 0.2 and 0.4 mm. At 0.8 mm it is worse (0.547 vs 0.526), because its accuracy degrades faster as the threshold loosens. Taking each model's best setting here would be chosen on the test set, so it is not reported as a result. This says nothing about the view-count threshold, MVSNet-style filters or the CL-MVSNet gap (see the box above).

Full output: runs/fusion_sweep_dist_n4/report.log. The full-grid results for scan1, scan4 and scan9 in runs/fusion_sweep/ are not used. Sweep finished 2026-10-11 20:35 UTC (~17–23 min per model-scan, far faster than the full grid).

## 4. Track 1 screen: frozen mono features (B2_mono_feat seed 2)

| Run (seed 2) | Acc | Comp | Overall |
|---|---|---|---|
| B2_nogeo (matched control) | 0.461 | 0.410 | 0.436 |
| B2_mono_feat | 0.464 | 0.382 | 0.423 |
| mono_feat − control | +0.003 | −0.029 | **−0.013** |

**Screening result, not a confirmed gain.**

**Matched setup (checked against configs and logs):**

- Both runs are initialised from **runs/B2_nogeo_finetune/seed2/ckpt_final.pt** (the "initialized weights from" line is in both train.log files). Everything upstream of the final stage is the same checkpoint.
- Same final-stage budget: 1500 steps each, lr 5e-4, same data, batch and loss settings. The config diff (B2_nogeo.yaml vs B2_mono_feat.yaml) shows only the mono-feature input (frozen Depth-Anything-V2-Small, zero-initialised injection at coarse and mid), data.geometry_prior (mono_sparse vs none) and use_geo_term (true vs false).
- **L_geo is off in both:** geo_weight 0.0 in both; the control has use_geo_term true with weight 0. B2 never reads the prior input, so the geometry_prior and use_geo_term differences do not change the computation.
- One budget difference: mono_feat has 12,480 more trainable parameters (2,133,939 vs 2,121,459, the injection layers), plus the frozen Depth-Anything model. Step time is about the same (4.17 vs 4.21 s/step).

**Result:**

- ΔOverall −0.0127 (mono_feat − control), larger than 0.009.
- Better on only 11 of 22 scans.
- scan48 alone accounts for −0.155 (0.870 → 0.715). Without scan48, the mean difference over the other 21 scans is −0.006, below 0.009.
- Single seed.
- The gain is from completeness (−0.029); accuracy is slightly worse (+0.003).

We read this as a screening signal concentrated on the ambiguous scan, not a confirmed gain. A second seed would be needed before Track 1 counts as a result.

**Per-scan Overall (mm), seed 2**

| Scan | Control (B2_nogeo) | B2_mono_feat | ΔAcc | ΔComp | ΔOverall |
|---|---|---|---|---|---|
| scan1 | 0.280 | 0.281 | +0.002 | +0.000 | +0.001 |
| scan4 | 0.335 | 0.343 | +0.008 | +0.008 | +0.008 |
| scan9 | 0.348 | 0.357 | +0.014 | +0.004 | +0.009 |
| scan10 | 0.298 | 0.303 | +0.009 | +0.001 | +0.005 |
| scan11 | 0.681 | 0.617 | +0.003 | −0.131 | −0.064 |
| scan12 | 0.339 | 0.342 | +0.011 | −0.006 | +0.003 |
| scan13 | 0.521 | 0.510 | +0.026 | −0.048 | −0.011 |
| scan15 | 0.323 | 0.334 | +0.017 | +0.006 | +0.011 |
| scan23 | 0.311 | 0.321 | +0.014 | +0.006 | +0.010 |
| scan24 | 0.325 | 0.344 | +0.015 | +0.022 | +0.019 |
| scan29 | 0.433 | 0.419 | −0.025 | −0.003 | −0.014 |
| scan32 | 0.499 | 0.442 | −0.004 | −0.109 | −0.057 |
| scan33 | 0.491 | 0.491 | +0.004 | −0.005 | −0.000 |
| scan34 | 0.243 | 0.239 | −0.004 | −0.004 | −0.004 |
| scan48 | 0.870 | 0.715 | −0.006 | −0.305 | **−0.155** |
| scan49 | 0.589 | 0.606 | +0.001 | +0.033 | +0.017 |
| scan62 | 0.570 | 0.543 | −0.027 | −0.027 | −0.027 |
| scan75 | 0.381 | 0.377 | +0.012 | −0.020 | −0.004 |
| scan77 | 0.740 | 0.732 | +0.012 | −0.028 | −0.008 |
| scan110 | 0.397 | 0.377 | −0.016 | −0.024 | −0.020 |
| scan114 | 0.261 | 0.264 | +0.003 | +0.003 | +0.003 |
| scan118 | 0.347 | 0.348 | +0.001 | −0.000 | +0.001 |
| **Mean (22)** | 0.436 | 0.423 | +0.003 | −0.029 | **−0.013** |

Δ = mono_feat − control; negative is better. mono_feat is better on 11 of 22 scans.

## 5. L_geo on/off: C1 vs B2_nogeo (full chain, L_geo off in every phase)

> **L_geo: n = 1, second seed pending.**

| Seed | Run | Acc | Comp | Overall | Per-scan Overall wins (of 22) |
|---|---|---|---|---|---|
| 2 | C1 (L_geo on, geo_weight 0.5) | 0.474 | 0.362 | 0.418 | 11 |
| | B2_nogeo (L_geo off) | 0.461 | 0.410 | 0.436 | 11 |
| | off − on | −0.013 | +0.048 | +0.018 | |
| 0 | C1 (L_geo on) | 0.470 | 0.358 | 0.414 | **PENDING** |
| | B2_nogeo (L_geo off) | **PENDING** | **PENDING** | **PENDING** | **PENDING** |
| | off − on | **PENDING** | **PENDING** | **PENDING** | |
| **Paired diff, mean ± std (ddof=0), n = 2** | | **PENDING** | **PENDING** | **PENDING** | same sign in both pairs: **PENDING** |

On seed 2, turning L_geo off makes overall 0.018 mm worse. That would mean L_geo helps, but the effect is mixed:

- All of it comes from completeness (+0.048).
- Accuracy is 0.013 better with L_geo off.
- The scan wins split 11 / 11.

With one seed we cannot tell this apart from seed-to-seed variation. C1's own seed spread is 0.414–0.430. The seed 0 pair decides whether "L_geo helps C1" can be claimed.

## 6. Not run, by decision

- C1 placebo prior (blurred, noisy and cross-scene G), configs prepared; patches/C1_placebo.patch not applied.
- No other GPU jobs until B2_nogeo seed 0 finishes.